

Giacomo Rossettini: "Placebo, nocebo and contextual effects: the missed

20:07

welcome to the fourth active inference

Symposium Dr gomo

rostini we're very pleased to have you

here and um looking forward to your

session titled PL ebo nosbo and

contextual effects the missing the

missed Link in clinical

practice over to

you thank you for the presentation and

for the invitation to use this talk it's

a pleasure for me to be here for the

hativ inference Symposium and today I

talk about clbo noo and contextual

effects showing the Mis Link in clinical

practice as I mentioned I'm an Italian

physical therapist I'm involved in

clinici in research and teaching session

I come from Verona that is a city in the

north of Italy and it represent the city

of love on Romeo and Juliet and if you

have never visited my suggestion is to

do

it before to start I want to declare

that I have no conflict of interest for

my

speech I Have organized my presentation

to discuss contextual effects firstly we

will analyze the definition and the

constituent dimensions of contextual

effects the mechanisms of action the

impact on outcomes and treatments next

we will discuss the possible link of

contextual effects with the active

inference and the implication for

clinical practice and research finally

we will conclude with a take on message contextual effects are defined as the effects of the context around the therapy and are generated by contextual factors contextual factors were recently defined in international consensus in which I participated in the slide you will find an accept of the final share definition where contextual effector are the perceived views that affect both the patient and the practitioner and can arise from previous experiences and the immediate Dynamics within the therapeutic encounter or a combination of both and from a clinical perspective contextual factors fall into five broad categories or Dimensions that you can see represented in this graph on one hand we have the characteristic of the clinicians with their professional reputations the appearance the beliefs and behaviors on the other side there are the patients characteristic with their expectations preferences and previous experiences between the clinicians and the patients there is the relational Dynamics in terms of verbal and nonverbal communication what we say and how we communicate in in addition to the relational component the treatment characteristics also play an important role they represent the rituality of the treatment finally The Fifth Dimension is represented by the characteristics of the setting within which the encounter

is being delivered such as the environment the architecture and also the interior design well all these five Dimensions interacting with each other influence the overall therapeutic outcomes the question we have to ask ourself is through which mechanisms does this happen I will tell you in the next slides contextual Factor have been studied for their ability to induce Placebo and noo effects which represent the effects of the psychosocial context around the th and to explain how this occurs I want to start with a clinical example this illustration show a common clinical encounter in which patients with the muscular skial pain receive a therapy such as joint immobilization therapeutic exercises or an ultrasound injections and during the administration of therapy patients are influenced by the specificity of the treatment but also by various different psychos social cues such as the clinicians work the Sight and Sound of the health care devices the smells of the hygienic and also the taste of the drugs when receiving for example a painkillers well all these cues can influence the patient's perception shaping the experience in a positive or in a negative way triggering Placebo or nobo effects thus the positive use of contextual factors can produce Placebo effects when these elements are used negatively it produce noo effects and the clinical effects of contextual factors can be explained through

conscious and unconscious mechanisms of action such as expectation learning processes personality traits mindset and even if genetics however among the most important psychological theories expectation classical conditioning and social learning are the most widely accepted the expectation is a conscious phenomena through which the patient expect that something will happen in the Futures we need to consider patients for example who are waiting to receive a therapeutic exercises or manual therapy for a muscular schal pain well they will predict a positive therapeutic outcome by evaluating all the contextual factors around the therapy we offer as a clinician such as our Worth or our behaviors instead the classical pav of condition is is an unconscious phenomenon concerning the past events in fact patients can learn over time to associate a certain therapy such as for example an inactions with a specific context effects that represent the rituality of the technique created by the needles that produce a certain outcome which can be positive or negative but patients can also learn from others they can learn from positive or negative experiences of other patients family members friends or from information they get from social media or blogs regarding for example muscular skial pain but how the contextual Factor Works in practice I will use this image to explain this mechanisms of action

briefly whenever patient interact will the healthcare context that are always influenced by the five contential factors and these elements are encoded by the patient brain who interprets them and provides an evaluation and a prediction of their meaning and if this interpretation is positive Place effects are triggered if the interpretation is negative noo effects are stimulated and these effects are sustained by a Cascade of a psychon immuno endocrinological events at multiple level of the central nervous systems in terms of the release neurotransmitters for example negative contextual factors can activate the release of kyin and the deactivation of dopamine and the opioid systems instead the use of the positive Context factors can facilitate the release of a PES endoc canids andam and also different brain areas can be activated for example the dorsal lateral peral cortex the anterior singulate cortex and the g per gray and the dorsal Horn of the spine are modulated and all these specific neurological Pathways translated into changes in clinical outcomes and using for example muscul scal pain as an example if the context oper is positive the patients will perceive a relief of symptoms instead if the Context adopted is negative the patient will have an aggravation of symptoms and contextual factors interacting with the treatment specificity are able to influence more the patient subjective outcomes or the

illness dimensions than the objective outcomes that represent the disease component for example pain disability experience satisfaction represent typical example of outcomes influenced by contextual factors however not all the changes of the therapeutic outcomes are due by contextual factors or by the specificity of our treatment in fact several Compounds can explain the improvements or the worsening of the patient clinical condition an example are reported in this graph for example the natural history of the disease can explain the natural fluctuation of the of symptom or the regression towards the means which represent a statistical phenomena or for example the presence of unidentified intervention as in the case of patient that present a muscular skeletal pain or receive a manual therapy or therapeutic exercise and concomitantly receive also a painkillers another question we should have ask ourself is what is the weight of contextual factors influences the outcome and for a long time it was a common opinion in the scientific Community that about 33 to 40% of the total therapeutic effects depend on the contextual component and even if it may seem limited it is actually not because it is a component influencing clinical outcome that we should know and we should use in our clinical practice and the international debate on the weight of contextual effects has continued to the present day

without find the conclusion

several recent systematic reviews with the meta-analysis has assessed the weight of contextual effects what the different authors have observed is that the contextual effects can vary from small to large and I report here different example you can see what happen for no specific loic pain neck pain fibromyalgia and also NEOS arthritis and the heterogeneity in these results can be explained for example by the characteristic of the patients the clinicians by the characteristic of the condition analyze the interventions considered and also by the characteristic of the statistical metric adopted so some student has also provided data on the contextual effects for treatments for example in this meta research the contextual effects of of different treatments was evaluated and what we can observe is that the contextual is progressively higher in treatment with the high realistic Futures to go into details in psychological treatments the the contextual effects account for the 38% in pharmacological treatments it accounts for the 61% and in the physical treatment such as manual therapy or handson techniques it accounts for the 65% 64% and having reached this point I think that we need to understand how contextual effects can be related to the theme of this coner this conference the active

inference and to understand this we need to start with a clinical example we need to think about the management of patients with persistent muscular skeletal pain who have negative expectations based on their previous experiences psychological characteristics or personality traits and for example if you consider a patient with the neck pain that in the past has been exposed to exercises and learn that these exercises produces pain consolidating a negative internal model to worth this in our patients Clinic we offer exercises to our neck pain patients in accordance with the best evidence and as a result of this direct experience the patient can have two main outcomes at first for example it can confirm that exercises can produce pain or the patients may not perceive the pain violated their expectations and this violation of expectation is interesting because it can result in two main scenarios one adapted IND where patients Mak an it makes an update of their expectation and on their internal model and a d functional one where the patients consider this violation as an exception to the rule or as results to the chance and therefore maintain it and this is referred to as a cognitively immuniz patients and we suggest that such differen in updating responses can be understood it at least to some extent from a basian perspectiv and in fact the ability to

make an update of the patient expectation can be interpreted in the light of the Bayesian model and will depend on the strength and accuracy of the patient prior beliefs and behaviors and from a Bayesian perspective our brain conceptualizes a predictor machine that generates predictions known as PRI about the expected sensory inputs and the integration between the prior and the sensory inputs results in the posterior the percept with which can be more or less influenced by the prior and by the sensory data depending on their level of precision and let us think about the patients with the pain who have negative prior or expectations toward exercise they represented the sensory input and if these priors are very precise and reliable it will be much more difficult for the patient to rely on the sensory input given by the exercises and the prediction error is really minimal and the patients consider the experience as a result of a change or an exception of the rule existing in maintenance or the initial expectations instead if the interpretation is less precise and reliable it will be much easier for the patients to rely on the sensory inputs given by the exercise because the prediction error between what it is expected and what it is experienced will be more significant will contribute to improving the patient expectation by generating update and I think is very

fascinating and complex but it does allow us to think about how we can use the context to challenge or make an update of the patient's prior and I will try to help you to understand this point with another clinical example with this B model in mind it is possible to better understand the clinical scenario presented in this image where we have the patient with the musculoskeletal pain of the shoulder has had a negative PR in that in that situation the patients prefer not to rise their harms because they have been told that the tendons can stop on the other hand we have the clinicians who tries to challenge the patient's prior through the tools that he has disposal for example the exercises the manual therapy the virtual reality and the education but the clinician also has another secret weapon to use that this scholar has the contextual factors made by words cues behaviors rituality that can influence the patient's brain and the patient's prior and as an outcome of this challenge we can have for example expectation violation following by an update which is realized when the prior with low Precision of our patient is updated accordingly with the newly acquired evidence the patients leave the brace the arms without any pain and without any types of lesions another possible scenario is represented by the expectation violation followed by the maintenance of the

dysfunctional beliefs as the consequence of one having a highly precise prior which is considered really reliable and therefore the newly acquired information and this confirmatory evidence is not sufficient to disprove and update the such strong prior and for example in this case we have a patient with a muscular crick pain that during a clinical session manage to leave their hand without any negative consequences to their shoulder but still believe that leaving the arm will possibly create and lead to the shoulder to break are likely to have for example a very high precise prior and might discard the positive evidence for example the success in lifting the hand which might be classified as an exception instead of the rule so try to pull the string of the speech what might be the implication for this uh from this talk to our clinical practice the use of contextual factors represent an opportunity to Aras the evidence based therapy we know it as a clinician but as a clinician we also need to consider that using context of factors such as words touch the rituality of the technique and environment could help to challenge the patients prior and improve the patient muscular crick pain however this observation that arise from the clinic need to be supported by data and more study to substantiate their effectiveness thus researchers are called up to integrate contextual

effects into the study on muscal pain
using for example the basian brain and
the active inference lens and to do is
this it is important to produce
experimental work in the lab in the
click work in patient to validate these
theoretical models by using for example
subjective outcomes but also some
objective physiological
metrics we are in the end so I concluded
I want to cre in you wishing you to
remain curious clinicians capable to see
the things from different perspective
and asset the complexity presented in
the clinic and also within this image
let's not just focus only on the details
the te the technique it is important but
it is not all let us always remember
that the scenario in which we operate it
is much more complex and both the
context the moon and the confounders The
Surge and the sky comes into play only
if we Embrace this opportunity and this
complexity we could find new strategy to
best manage our patients with muscular
skial pain thanks for your
attentions thank you
thank you till next time